AP Physics C Study Guide

Mr. Maine, North Hollywood High
Based on lectures taught from 8/12/26 to x/x/27

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Misc. Notes

1. This study guide assumes a knowledge of simple calculus (mostly just derivative + integral rules) and vector math.
2. When it's more convenient, I use the dot notation for derivatives, such that $\dot{x} = \frac{dx}{dt}$ and $\dot{\vec{r}} = \frac{d\vec{r}}{dt}$, because it's faster to type.
3. PLEASE let me know if I made a mistake or missed something. I'm sure I did.

Units and Dimensional Analysis

Units

Almost everything has units, and they matter. When you write a variable, the units are built into the variable, e.g. $a = 15\,\frac{m}{s^2}$ or $\vec{v} = \langle 10\frac{m}{s}, 12\frac{m}{s}, -4\frac{m}{s}\rangle$.

Unit Systems

For this class, we mostly use the SI (*Système international d'unités* AKA International System of Units) system – the basic units used in AP Physics C are meters(m), kilograms(kg), seconds(s), and amperes (A). The other base units of SI are the kelvin (K), the mole (mol), and the candela (cd), but these aren't used in this course. Derived units, like the Newton or the Watt, are built from these basic units.

Prefixes, like kilo-, centi-, or milli-, can be prepended to a unit to multiply it by a specific factor.

The other metric system is the CGS (centimeters, grams, seconds) system, which can be useful for smaller values.

The AP test uses the SI system only, and Mr. Maine will mostly use SI. When Maine uses units from another unit system, he will tell you about all relevant conversions.

Dimensional Consistency

Dimensional consistency is always necessary. Here are its requirements:

1. Any two quantities that we equate must have the same dimensions.
   Correct: $(3m)(2m) = 6m^2$
   Incorrect (obviously in this case, but with lots of operations it can be hard to tell): $5.3g = 2\,\frac{m}{s}$
   Because units can be consistent without the same unit system being used, e.g. $1m = 3.28ft$, we often use generic units when doing dimensional analysis:
   M: Generic unit of mass
   L: Generic unit of length
   T: Generic unit of time

2. If you're adding or subtracting quantities, they must have the same dimensions.

3. Dimensions obey algebra rules.
   For example, $\Delta x = v_0 t + \frac{1}{2}at^2$ in generic dimensional form is $L - L = \frac{L}{T} * T + \frac{L}{T^2} * T^2 \rightarrow L = L$

Kinematics

Position

Position can be any variable, but it's common to default to x, or, as a vector, $\vec{r}$.

Speed and Velocity

Speed and velocity are two different quantities: speed is the magnitude of velocity (i.e. how fast is an object going), whereas velocity also has direction (i.e. how fast an object is going and in what direction).

With velocity as a scalar:

$$v = \dot{x}$$

$$speed = |v|$$

And as a vector:

$$\vec{v} = \dot{\vec{r}}$$

$$speed = ||\vec{v}||$$

Acceleration

Just the rate of change of speed:

$$a = \dot{v} = \ddot{x}$$

Or as a vector:

$$\vec{a} = \dot{\vec{v}} = \ddot{\vec{x}}$$

Equations Using Constant Acceleration

These equations are only true when acceleration is CONSTANT. Do not use otherwise.

In this section, I'll write position, velocity, and acceleration as scalars, as one would do for a 1D motion problem. In 2D or 3D problems, it's useful to split everything up into individual components and solve each component like an independent 1D problem.

$$\Delta x = v_0 t + \frac{1}{2} a t^2$$

$$\Delta v = at$$

$$v_{avg} = \frac{v_0 + v}{2}$$

$$v^2 = v_0^2 + 2a\Delta x \rightarrow a \propto \frac{1}{\Delta x}$$

Note: you can use $v_{avg} = \frac{\Delta x}{t}$ and $a_{avg} = \frac{\Delta v}{t}$ even when acceleration isn't constant.

Reference Frames

Reference frames are arbitrary. In "lab frame", which is the reference frame where walls, furniture, etc. are assumed to have $v = 0$, my computer's velocity is 0. W/r/t the sun? Its velocity is $\sim 30,000 \frac{m}{s}$!

Fortunately, we assume physics scenarios are in lab frame, unless otherwise specified. If we want to discuss reference frames, we introduce a new vector notation: $\overrightarrow{r_{A/B}}$, where A and B are both points, is the position of A w/r/t B, i.e. $\vec{a}$ given that the origin is at B. You can formally change reference frames like so:

$$\overrightarrow{r_{C/B}} + \overrightarrow{r_{A/C}} = \overrightarrow{r_{A/B}}$$

$$\overrightarrow{r_{V/P}} + \overrightarrow{r_{X/V}} = \overrightarrow{r_{X/P}}$$

It's best to not think about these formulas, though. Common sense and imagining viewing the same thing from two different perspectives are usually easier and faster.

The Range Formula

This formula is very limited. Scenario: a projectile is launched at a known angle from the horizontal, and lands at the same height. The projectile must experience no air resistance, and doesn't go far enough for the curvature of the earth or any variation in gravity to matter. Where does it land?

$$\Delta x = \frac{v_0^2 \sin 2\theta}{g}$$

We haven't seen g yet, but it's simply the acceleration due to gravity when an object is under a planet's gravitational pull. On Earth, g approximates to $9.81 \frac{m}{s}$. On the AP exam (and on Maine's tests) you can use $g = 10 \frac{m}{s}$.

Forces, Dynamics, and Newton's Laws

Definition of a Force

A force is, at its core, a push or a pull.

In SI units, the unit for a force is the "newton", or "N".

$$1N = 1\frac{kg * m}{s^2}$$

There are four fundamental forces: gravity, strong nuclear force, weak nuclear force, and electromagnetic force. In this class, we only really care about gravity and electromagnetic force. Most of the forces we see (friction, tension, normal force, drag, applied force, spring/elastic force, etc.) are just the complicated sums of a lot of EM interactions.

Newton's Laws

Newton's 3 laws relate force to motion:

1. In the absence of external forces (or if the sum of external forces is 0), an object will maintain constant velocity.
2. $\sum \overrightarrow{F_{ext}} = m\vec{a}$
   That's the version we'll use, but the actual version is defined in terms of momentum: $\overrightarrow{F_{ext}} = \dot{\vec{p}} = m\vec{a} + \vec{v}\dot{m}$
3. If an object A exerts a force upon object B, then object B exerts an equal and opposite force on object A.

Force of Gravity

The force exerted by gravity is also known as the "weight" of an object. This is important – if a test problem asks for an object's weight, don't answer with its mass, answer with the force exerted on it by gravity.

$$\overrightarrow{F_g} = \vec{w} = m\vec{g}$$

Normal Force

We subdivide the contact force between two objects into two forces: normal force ($\overrightarrow{F_N}$), which acts in the perpendicular direction and prevents illegal acceleration (i.e. an object clipping into another object), and friction ($\overrightarrow{F_F}$), which acts in the parallel direction and which we'll talk about later.

Gravitational Forces and Fields

Newton's Law of Universal Gravitation: For two point masses M_1 and M_2, separated by a distance d, the attractive force has magnitude

$$||\overrightarrow{F_g}|| = \frac{GM_1M_2}{d^2}$$

G is the universal gravitational constant. It equals $6.674 * 10^{-11}\frac{Nm^2}{kg^2}$.

$$\overrightarrow{F_{g1on2}} = \frac{GM_1M_2}{r^2}\hat{r_{2/1}}$$

This gives us a new perspective on $\vec{g}$: It's $\frac{\overrightarrow{F_g}}{m}$, so we can also say that the units of $\vec{g}$ are $\frac{N}{kg}$ as well as $\frac{m}{s^2}$.
$\vec{g}$ can also be called the gravitational field.
In the vicinity of a mass M, the gravitational field is

$$||\vec{g}|| = \frac{||\overrightarrow{F_g}||}{m} = \frac{GM}{r^2}$$

Note that this does not depend on the mass of the object gravity is acting on.

Shell Theorem

A thin, uniform spherical shell of mass exerts the same force on an outside object as though it were a point mass located at its center. Additionally, it exerts no force on an object inside the shell. Given that, for our purposes, planets can be approximated as several thin, nested shells, this is very useful.

A useful application of the shell theorem is this, where $\overrightarrow{g(r)}$ is the gravitational field and r is the distance of a point from the center of a large, spherical, uniformly dense mass M of radius R:

$$\overrightarrow{g(r)} = \begin{cases} \frac{GM}{R^2}\left(\frac{r}{R}\right), & r \leq R, \\ \frac{GM}{r^2}, & r > R. \end{cases}$$

Friction

Static Friction

Static friction prevents slipping between solids, up to a certain limit. There is no equation for the force of static friction. Up to a limit, it's whatever prevents slipping. The limit has an equation:

$$||\overrightarrow{F_{F,maximum}}|| = \mu_s||\overrightarrow{F_N}||$$

Kinetic Friction

Kinetic friction opposes motion when there is relative motion/slipping between solids:

$$||\overrightarrow{F_{F,kinetic}}|| = \mu_k||\overrightarrow{F_N}||$$

μ is the coefficient of friction, which depends on the two solids in question. The coefficient of static friction is always larger than the coefficient of kinetic friction.

Springs, Elasticity, and Hooke's Law

For many objects, length change is proportional to applied force. Hooke's law describes that length change ΔL:

$$||\overrightarrow{F_{spring}}|| \propto ||\Delta L||$$

The direction of $\overrightarrow{F_{spring}}$ is always "restoring" i.e. pointing back toward the relaxed length. The proportionality constant is usually called "k" or "spring constant"

$$||\overrightarrow{F_{spring}}|| = k||\Delta L||$$

We always determine k through experiment. It quantifies the stiffness of a spring, where a small k denotes a not-stiff spring, like a slinky, and a large k denotes a stiff spring, like a car shock.

The SI units of k are $\frac{N}{m}$ or $\frac{kg}{s^2}$.

Combining Springs

n springs in series:

$$k_{combined,series} = \frac{1}{\frac{1}{k_0} + \frac{1}{k_1} + ... + \frac{1}{k_n}}$$

n springs in parallel:

$$k_{combined..,parallel} = k_0 + k_1 + ... + k_n$$

Circular Motion

Circular motion is when an object moves along the arc of a circle. Think of a turning car, or a toy being swung in a circular arc by a string.

In this situation, an object's net force will result in an acceleration, called the "centripetal acceleration", pointing towards the center of the circle. The magnitude of that acceleration is described by this formula:

$$a_c = \frac{v^2}{R}$$

Where v is the object's speed and R is the distance between the object and the center of the circle whose arc the object is tracing, or

$$a_c = \omega^2 R$$

Where ω is the objects angular speed, in radians per second.

When an object is experiencing circular motion and not speeding up or slowing down, which is called Uniform Circular Motion (UCM), the entirety of their acceleration is towards the center of their circle (i.e. the object's $a = a_c = \frac{v^2}{R}$).

When an object is experiencing Non-Uniform Circular Motion, which means that they're either speeding up or slowing down while turning, a_c is just one component of their acceleration, the one pointing towards the center of their circle.

It is extremely important to note that an object experiencing circular motion is not experiencing any new forces. An object's experiencing circular motion simply means that the net force exerted on them consistently points toward the center of a circle.

Center of Mass

Intuitively, the center of mass (CoM) of a system is the weighted average position of all the individual masses in the system.

In simple objects, the CoM is obvious by symmetry. The CoM needs to lie on any mirror planes in an object.

For complex objects, the CoM is usally found by experiment – if you balance or hang an object from a point, the CoM will be directly above or below the point in question.

There is a mathematical way to find the CoM. For a collection of point masses $m_1, m_2, ..., m_n$ at positions $\overrightarrow{r_1}, \overrightarrow{r_2}, ..., \overrightarrow{r_n}$

$$\overrightarrow{r_{CoM}} = \frac{\sum_{i=0}^{n} m_i \overrightarrow{r_i}}{\sum_{i=0}^{n} m_i}$$

With a solid that isn't easily modeled with finitely many point masses:

$$\overrightarrow{r_{com}} = \frac{\int \vec{r} dm}{M}$$

or

$$\overrightarrow{r_{com}} = \frac{\int \vec{r} (\rho dV)}{M}$$

Orbital Mechanics Pt. 1

We'll revisit orbits later, when we know more about energy and momentum, but for now we're restricted to very simple circular orbits.

First, the simplest scenario: a circular orbit where one object is much larger than the other.

If $M >> m$, we can figure that the CoM of the system is basically the center of the larger object, and that the larger object is basically still.

T is the orbital period, or the time to go around the orbit once; v_{orbit} is the speed of the smaller object; M is the mass of the larger object, and r is the distance between the centers of the two objects.

$$v_{orbit} = \sqrt{\frac{GM}{r}}$$

$$T = \frac{2\pi r}{v_{orbit}}$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$T^2 \propto r^3$$

So, if we compare two bodies m_1 and m_2 orbiting the same M with orbital radii r_1 and r_2, and orbital period T_1 and T_2, respectively:

$$(\frac{T_1}{T_2})^2 = (\frac{r_1}{r_2})^3$$

If M is not significantly larger than m, like in the case of Pluto and its moon Charon (the mass of Pluto is only about four times the mass of Charon), then they both orbit around their CoM. Two applications: they will both always be on opposite sides of their CoM, which will cause the CoM to always remain in the same place; and their angular velocities will be the same regardless of their linear velocities.

Work, Energy, and Power

Energy

Energy is a scalar quantity.

It's conserved, which means it can't be created or destroyed; it can be moved around, or converted between forms. It always needs to come from or go somewhere.

Kinetic energy is the simplest kind of energy to define: It's energy associated with an object in motion.

For a point of mass m moving at speed v, the kinetic energy K is

$$K = \frac{1}{2}mv^2$$

K depends on reference frame, so when working with energy it is extremely useful to not switch reference frames mid-problem.

Work

Simplest definition of work: if an object moving along the x-axis experiences a constant force along that axis, and undergoes a displacement Δx, we say the force does work:

$$W = F\Delta x$$

Work is signed. It's positive if both F and Δx are parallel, and negative of F and Δx are antiparallel.

If we leave one dimension, the work exerted on the object by $\overrightarrow{F}$ is the object's displacement multiplied by the component of $\overrightarrow{F}$ that is parallel to the object's motion, so

$$W = \overrightarrow{F} \cdot \Delta\vec{r}$$

If a non-constant $F_x(x)$ (Where F_x is the component of $\overrightarrow{\mathrm{F(x)}}$ parallel to the object's displacement) acts over a displacement from x_1 to x_2:

$$W = \int_{x_1}^{x_2} F_x(x)dx$$

In the general case, where an object moves along an irregular curve while a non-constant $\overrightarrow{F}(\vec{r})$ is being applied to it:

$$W = \int_{path} \overrightarrow{F}(\vec{r}) \cdot d\vec{r}$$

Work-Kinetic Energy Theorem

Work is tied to the net kinetic energy:

$$W = \Delta K$$

Conservative and Nonconservative Forces

Work done by a **conservative force** depends only on starting and ending positions, not the path taken between them. As such, they're also called "path-independent" forces. All conservative forces have an associated potential energy.

Examples of conservative forces:

- gravity
- ideal springs

Work done by a **nonconservative force** does depend on path and is not associated with a potential energy.

Examples of nonconservative forces:

- friction
- drag
- tension
- normal force

Potential Energy

Potential energy in a system is energy associated with the relative positions of objects in the system rather than their speeds. We can think of it as stored energy.

For any conservative force F,

$$W_F = -\Delta U$$

Conventionally, U is the symbol for potential energy.

As such, given that $W_F = \int_{path} \vec{F}(\vec{r}) \cdot d\vec{r}$,

$$dW = -dU \rightarrow \vec{F} \cdot d\vec{r} = -dU \rightarrow \vec{F} = \vec{\nabla} U$$

Don't worry! We'll be working almost entirely in 1D, where

$$dW = -dU \rightarrow F_x dx = -dU \rightarrow F_x = \frac{-dU}{dx}$$

So just derive $U(x)$ to find the associated force. But given $F(x)$, there are infinitely many possible associated potential energies because of the $+C$ we get when we integrate. Let's look at gravity on an object, for which $F_g = -mg$. So

$$U(x) = -\int F_x dx = mgx + C$$

The C, fortunately, is arbitrary. We usually pick it such that $U = 0$ at a convenient location like the floor height.

How about for actual gravity?

$$U(x) = -\int -\frac{GM_1M_2}{x^2} dx = -\frac{GM_1M_2}{x} + C$$

This graph looks like the graph of $\frac{1}{x}$ flipped over. C should be chosen so that the horizontal asymptote is at 0, i.e. gravitational potential energy is 0 when the objects are infinitely far apart.

Another useful potential energy to know is that of a Hookean Spring, for which $F(x) = -kx$.

$$U(x) = -\int F(x) dx = \frac{1}{2}kx^2 + C$$

In this case, C has an obvious choice: 0. That way, potential energy is 0 when the spring is totally relaxed (when $x = 0$).

Equilibrium

Equilibrium is the position/state where the net force on an object is 0. In the case of potential energy, we can think of equilibrium as the state where the conservative force to which the potential energy is tied is 0. If we look at a graph of potential energy $U(x)$, the positions of equilibrium will be at critical points (i.e. minimums or maximums, consider the fact that $F(x) = -\frac{dU}{dx}$)). There are three types of equilibria:

- **Stable Equilibrium**, where moving the position of the object in question in either direction will result in a force which acts to return the object to equilibrium. On a graph of $U(x)$, this is a local minimum. Moving the object in the negative x direction, we reach a point on the graph of $U(x)$ where $F(x) = -\frac{dU}{dx}$) is pointing in the positive direction, and moving the object in the positive x direction, we reach a point where $F(x) = -\frac{dU}{dx}$ is pointing in the negative direction, both of which will work to move the object back to its stable equilibrium position.
- **Unstable Equilibrium**, where moving the object in either direction will result in a force which acts to move the object away from the unstable equilibrium point. Consider a local maximum.
- **Metastable Equilibrium**, where moving the object in one direction will result in a force which acts to move the object back to the equilibrium position, and moving it in the other direction will result in a force which acts to move it away from the equilibrium position.

Power

Power is the rate at which work is done, represented by the variable P:

$$P = \frac{dW}{dt}$$

The SI unit of power is the Watt, which is derived from the formula: one Watt $= \frac{J}{s} = \frac{1kg \cdot m^2}{s^3}$.

The formula for power gives us a few useful facts:

- $W = \int_{t_1}^{t_2} P dt$
- When P is constant, $P\Delta t = W$ and $P = \frac{W}{\Delta t}$.
- Whether or not P is constant, $P_{average} = \frac{W}{\Delta t}$ and $W = P_{average}\Delta t$
- $\vec{F} \cdot \vec{v} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \frac{\overrightarrow{F \cdot dr}}{dt} = \frac{dW}{dt} = P$